W13 · EMNLP 2026 WORKSHOPS
ArabicNLP 2026 · Shared Tasks
The Fourth Arabic Natural Language Processing Conference

› ALEXIS AT DIALECTSENTEVAL SUBTASK 1

SHARED TASK DialectSentEval 2026 SUBTASK 1

Adversarial Training and Iterative Self-Training for Zero-Shot Dialect Sentiment Analysis

1st OF 12 TEAMS

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0.8667 Macro F1 — official test set, rank 1 / 12
+4.6 pts Ahead of runner-up — 0.8667 vs 0.8210 Macro F1
0.8857 Accuracy — official test set
52.0% Lebanese OOV word types — a dialect with zero training rows

01 › THE TASK

Three-way sentiment of dialectal hotel reviews

Classify MDS-3 hotel-review sentences as **positive**, **negative** or **neutral**. Training covers four dialects; the released test set adds a fifth, **Lebanese**, with no labeled data at all.

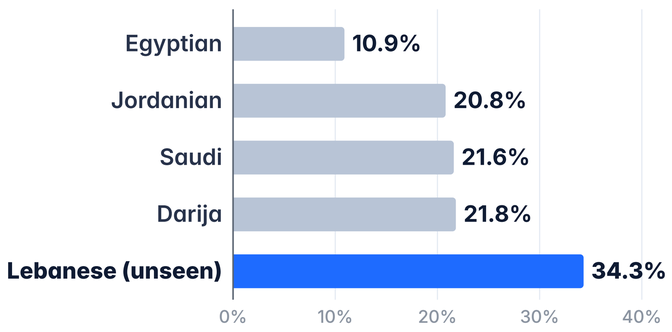
DIALECT	TRAIN	TEST
Darija (Moroccan)	433	105
Egyptian	432	105
Jordanian	433	105
Saudi	433	105
Lebanese (unseen)	**0**	**105**
Total	**1,731**	**525**

38.0% negative · **35.2%** positive · **26.9%** neutral

Classes are near-balanced and near-uniform across dialects, so dialect-wise gaps reflect genuine transfer difficulty, not label shift.

02 › THE DIAGNOSIS

A zero-shot transfer problem, not a classification problem



Out-of-vocabulary token-occurrence rate by test dialect. Lebanese reaches 34.3%, more than 1.5× any seen dialect. Over word types Lebanese reaches **52.0%**, vs 19.5–41.0% for the seen dialects.

- Negation markers appear in 44.4% of negative but also 37.8% of neutral sentences, so surface cues separate classes poorly. We use a **contextual transformer**, not a lexicon or rules.
- Median length is 16–19 words, so a **128 sub-word** cap discards almost nothing.

03 › VALIDATION PROTOCOL

k-fold CV cannot see an unseen dialect

k-fold measures generalization to unseen *sentences* from dialects the model has already seen. We add **Leave-One-Dialect-Out (LODO)** as the primary diagnostic and score every component on the official leaderboard.

	DAR.	EGY.	JOR.	SAU.	LEB.
LODO 1	test	train	train	train	–
LODO 2	train	test	train	train	–
LODO 3	train	train	test	train	–
LODO 4	train	train	train	test	–
Official	train	train	train	train	test

Each LODO fold trains on three seen dialects and tests on the fourth, simulating the official condition (amber): train on all four, test on unseen Lebanese. Alongside it we ran 5-fold StratifiedGroupKFold, grouped by exact text so the 17 duplicated training sentences never straddle folds.

04 › SYSTEM

Each component targets the diagnosed failure

1 Three dialectally diverse encoders

- **MARBERTv2** — strongest on AHASIS pilot
- **CAMeLBERT-DA** — dialectal pretraining
- **AraBERTv2** — MSA-centric, errs differently

Masked-mean pooling (unfamiliar sub-words cannot dominate one [CLS] point) and class-weighted cross-entropy.

↓

2 FGM adversarial training LARGEST SINGLE GAIN
Each step adds a bounded perturbation (ε = 1.0) to the word embeddings along the loss gradient and back-propagates through clean and perturbed passes: a proxy for moving from a seen dialect's embedding neighborhood to an unseen one's.

↓

3 Iterative self-training on the unlabeled test set
3 rounds: a committee of the current best models labels the test set; agreed rows above confidence 0.55 join training at 0.7× loss weight. Round 3 adds a **per-class** filter trimming each class's least-confident 10%, because a flat threshold under-represents neutral.
Coverage: 386 → 499 of 525 test rows.

↓

4 Character-level noise augmentation
One noised copy of every gold row: Arabic letter-variant swaps (ة/ه, ى/ي) plus delete, duplicate and swap at ≈6% per character. Targets the Lebanese OOV gap with no assumption about Lebanese class proportions.

↓

5 Uniform softmax ensemble + overlap control
Probabilities of the three backbones are averaged. The 58 test rows (11.0%) byte-identical to training sentences are a data-quality artifact: an exact-match lookup returns their gold label, so the models and ablations are judged on the 89.0% that need generalization.

05 › ERROR ANALYSIS · LODO

Neutral is the hardest class to recover

	NEG	NEU	POS
Jordanian	0.988	0.970	0.983
Saudi	0.976	0.950	0.984
Darija	0.911	0.888	0.933
Egyptian	0.928	0.856	0.914

Per-class F1 when each seen dialect is held out. Jordanian, Lebanese's Levantine relative, is easiest (F1 = 0.980); Egyptian is hardest (0.899), so dialect family alone does not fully predict difficulty.

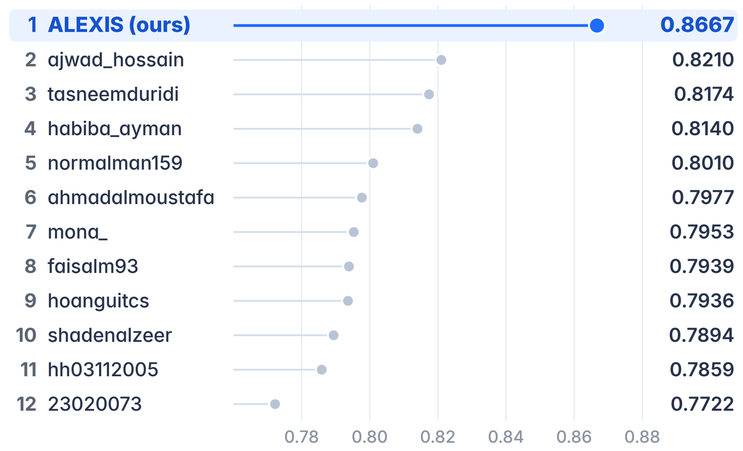
0.980 easiest held-out dialect: Jordanian (mean F1), the Levantine relative of Lebanese

0.899 hardest held-out dialect: Egyptian (mean F1)

AN OPTIMISTIC PROXY
LODO still draws held-out rows from the same translated-review pipeline as training, whereas Lebanese is genuinely unseen. Seen-dialect CV exceeds **0.93** Macro F1; the official score is **0.8667**.

06 › OFFICIAL RESULTS

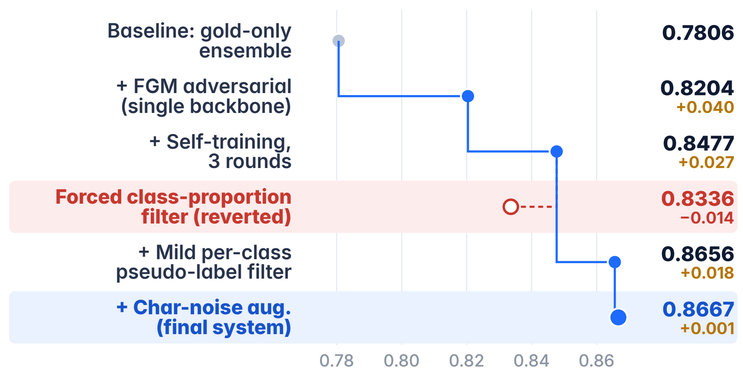
1st of 12 teams, 4.6 points clear



Official Subtask 1 leaderboard (Macro F1). Our submission also reaches 0.8857 accuracy. Submission IDs omitted.

07 › ABLATION

Every step scored on the real leaderboard



Cumulative deployed-system ablation (Macro F1), each row an independent leaderboard submission. The red row is the aggressive filter we reverted; the next step builds on 0.8477.

AN INSTRUCTIVE FAILURE
Forcing pseudo-label class proportions to match the four seen dialects regressed (**0.8336** vs 0.8477). It assumed Lebanese shares the seen dialects' distribution; the milder, dialect-agnostic filter does not.

TRIED, NOT DEPLOYED
R-Drop, SWAD, IRM, CORAL, SupCon, MixStyle, char-CNN/BiLSTM: **0.785–0.807**. Fish collapsed to **0.3215**; ByT5 reached 0.376 in 5-fold CV (never submitted). None beat FGM's 0.8204, consistent with our diagnosis: they target generic regularization or domain invariance, not the OOV mechanism.

08 › REPRODUCIBILITY

Key hyperparameters

SETTING	VALUE
Max sequence length	128 sub-words
Pooling / loss	masked mean / weighted CE
Class weights neg/pos/neu	≈ 0.88 / 0.95 / 1.24
FGM perturbation norm ε	1.0
Self-training rounds · conf.	3 · 0.55
Pseudo-label loss weight	0.7×
Char-noise rate	≈ 6% per char
Ensemble	uniform softmax avg.

TAKE-HOME MESSAGE

Diagnose the failure mode first. With an unseen dialect at test time, **validate leave-one-dialect-out** and choose interventions that match the **lexical and orthographic shift** (FGM and character noise), plus self-training on the unlabeled test set. Seen-dialect CV stays above 0.93 Macro F1 against 0.8667 officially: Lebanese is still the dominant error source, and closing it will likely need genuine Lebanese-domain signal.

PAPER
ACL Anthology 2026.arabicnlp-sharedtasks.187

CODE
github.com/Momen-Ibrahim-Fawzy/ALEXIS-DialectSentEval-Subtask1-SentimentAnalysis-

KEY REFERENCES
Ezzini et al. (2026). DialectSentEval 2026: Arabic Dialect Sentiment Analysis and Swapping Shared Task. ArabicNLP. · Alharbi et al. (2025). ADOR: Dataset for Arabic dialects in hotel reviews (MDS-3 data). · Miyato et al. (2017). Adversarial training methods for semi-supervised text classification. ICLR. · Srivastava & Chiang (2023). Fine-tuning BERT with character-level noise for zero-shot transfer to dialects and closely-related languages. VarDial. · Abdul-Mageed et al. (2021) MARBERT · Inoue et al. (2021) CAMeLBERT · Antoun et al. (2020) AraBERT.

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